Program : **Diploma in Robotic Process Automation**	
Course Code : **6309B**	Course Title: **Cloud Computing Fundamentals Lab**
Semester: **6**	Credits: **1.5**
Course Category: **Program Elective Course**	
Periods per week: **3 (L: 0 T: 0 P: 3)**	Periods per semester: **45**

Course Objectives:

- To install cloud computing environments
- To setup and configure VMs of different flavors of Linux and Windows
- To use resources in public cloud

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Programming concepts	3307	Problem solving using Python Lab	III
Programming concepts	**3136**	Database Management System Lab	III

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate to launch VM instances on Local Machine.	12	Applying
CO2	Illustrate to launch VM instances on cloud	6	Applying

CO3	Use virtualization products like VMWare/Virtual Box to create and run multiple virtual machines on an OS	9	Applying
CO4	Implement Single Node and Multi-Node Hadoop clusters.	12	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	**Demonstrate to launch VM instances on Local Machine.**		
M1.01	Demonstrate to Install VirtualBox/VMWare	2	Applying
M1.02	Illustrate to Configure VM instances of different flavors of Linux and Windows. Allocate CPU, memory and storage space as per a specified requirement.	5	Applying
M1.03	Experiment Launched VM instances to confirm the successful installation of the OS by performing a few OS commands.	2	Applying

M1.04	Make use of created virtual machines through Virtual Box to Install a C compiler and execute Simple Programs	3	Applying
CO2	**Illustrate to launch VM instances on cloud**		
M2.01	Demonstrate the Configuration of a VM instance in cloud by creating a cloud account	3	Applying
M2.02	Illustrate a procedure to launch virtual machine using trystack (Online Openstack Demo Version)	3	Applying
	Lab Test – I	3	
CO3	**Use virtualization products like VMWare/Virtual Box to create and run multiple virtual machines on an OS**		
M3.01	Demonstrate a procedure to transfer the files from one virtual machine to another virtual machine	6	Applying
M3.02	Implement Para-Virtualization using VM Ware's Workstation/ Oracle's Virtual Box and Guest O.S.	3	Applying
C04	**Implement Single Node and Multi-Node Hadoop clusters.**		
M4.01	Demonstrate the Installation and Configuration of single node Hadoop cluster on a VM	4	Applying
M4.02	Illustrate the Installation and Configuration of Multi-Node Hadoop Cluster on multiple VMs in same machine	4	Applying
M4.03	Experiment the Installation and Configuration of Multi-Node Hadoop Cluster on multiple VMs spanned across machines	4	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
R1	Erl, Thomas, Ricardo Puttini, and Zaigham Mahmood. Cloud computing: concepts, technology, & architecture. Pearson Education, 2013.
R2	White, T. (2009). Hadoop the definitive guide. O'Reilly Media. Sebastopol, Calif.
R3	Grover, M., &Malaska, T. (2015). Hadoop application architectures. O'Reilly.

Online Resources

Sl.No	Website Link
1	https://cloudacademy.com/library/cloud-fundamentals/labs/
2	https://www.virtualbox.org/manual/ch02.html
3	https://kb.vmware.com/s/article/2053973
4	https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-common/SingleCluster.html
5	https://www.tutorialspoint.com/hadoop/hadoop_multi_node_cluster.htm